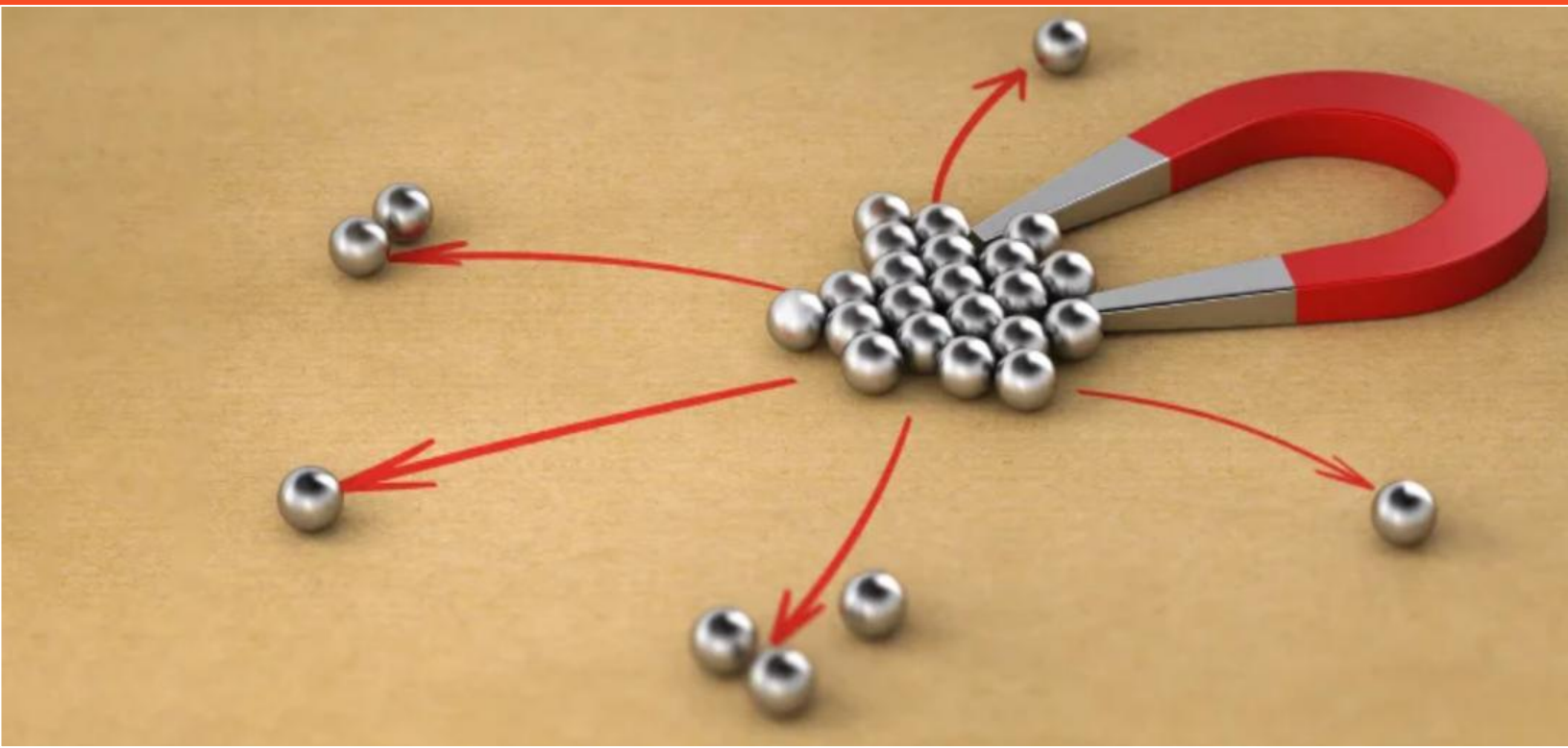




STRATEGIC ANALYSIS &
PREDICTIVE MODELING FINAL
REPORT:

PROJECT CHURNGUARD

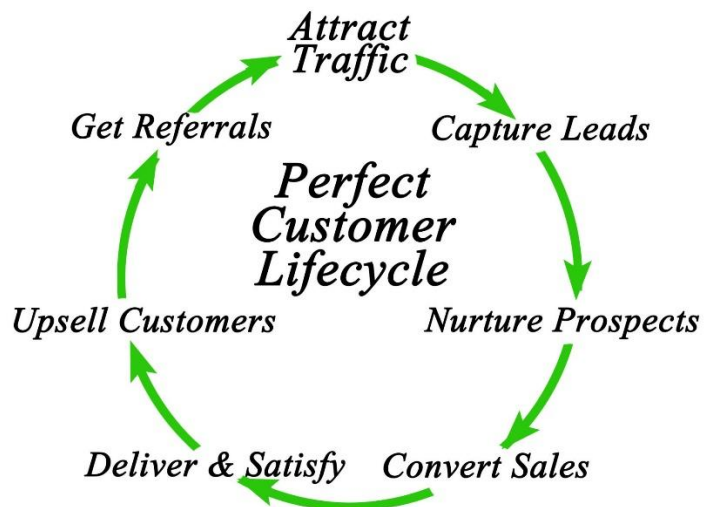
This report details the foundational analysis and predictive modeling of our 11,260-account database. Our primary objective was to move from reactive troubleshooting to a proactive, AI-driven retention framework to stabilize the 16.84% baseline churn rate.

By Sivin Mundayathil Stephen- Sr. Business Analyst.

1. EXECUTIVE SUMMARY

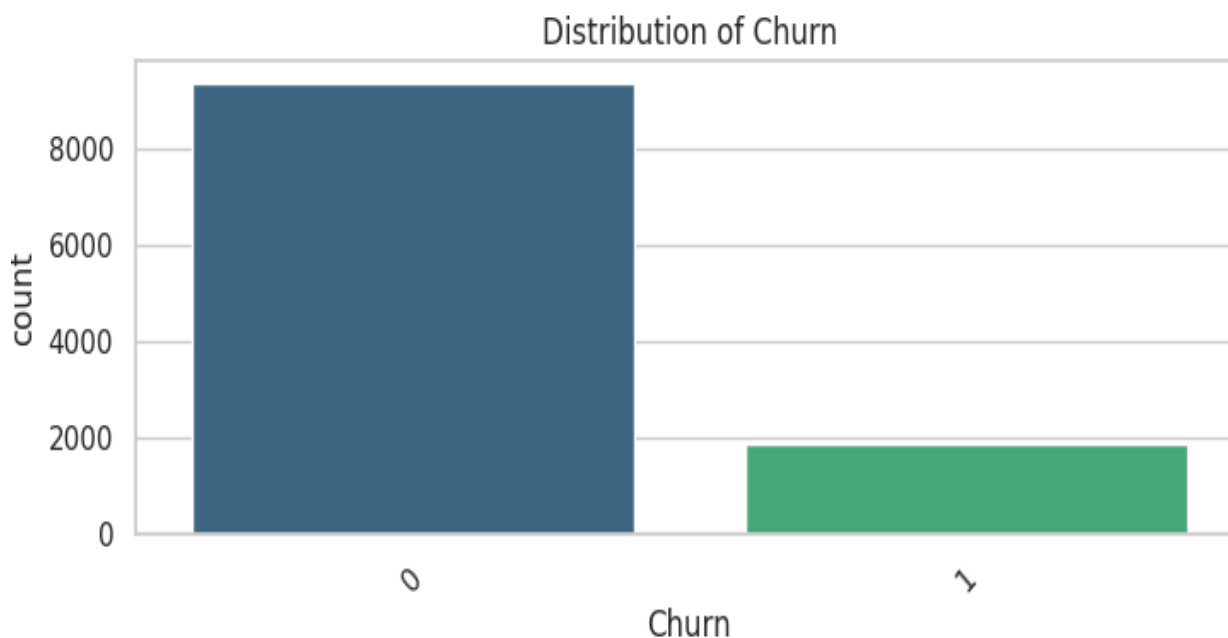
This report details the foundational analysis and predictive modeling of our 11,260-account database. Our primary objective was to move from reactive troubleshooting to a proactive, AI-driven retention framework to stabilize the 16.84% baseline churn rate.

Through extensive Data Exploration (EDA) and Machine Learning, we have identified that churn is not random. It is concentrated in the "Regular Plus" segment and the "Month 0" onboarding phase. By deploying our optimizedXGBoost predictive model, we can now identify 87% of potential churners before they leave.



2. BUSINESS CONTEXT: THE COST OF ATTRITION

To understand why Project ChurnGuard is a financial necessity, we must quantify the "Invisible Leak" in our revenue. Attrition affects the company in three distinct ways:



A. Negative ROI on Acquisition (The "Sunk Cost" Trap)

- Every account in our database was acquired through marketing spend, referral bonuses, or sales commissions (Customer Acquisition Cost, or CAC).
- When a customer churns within the "Red Zone" (Month 0-3), they leave before the company has broken even on that initial investment.
- High attrition in early tenure essentially means the company is subsidizing the customer's short-term usage without ever seeing a return.

B. Erosion of Customer Lifetime Value (CLV)

- Our ecosystem thrives on long-term "stickiness." A "Regular" customer might only spend 5.0 units monthly, but over 3 years (36 months), that account is worth 180 units.
- The Stability Dividend: As shown in our EDA, once a customer hits the 20-month mark, their probability of leaving drops to near zero.
- Every churn event represents the loss of that "Future Revenue" which is significantly more profitable because it requires zero additional marketing spend to maintain.

C. Operational Tax (The Support Burden)

- The data highlights a direct link between churn and high service friction.
- High-churn customers are often our most expensive to serve. Those reaching the "Breaking Point" of 25+ calls consume a disproportionate amount of staff time and technical resources.
- By predicting and resolving these issues early, we don't just save the customer; we reduce the Operational Overhead of the support department.

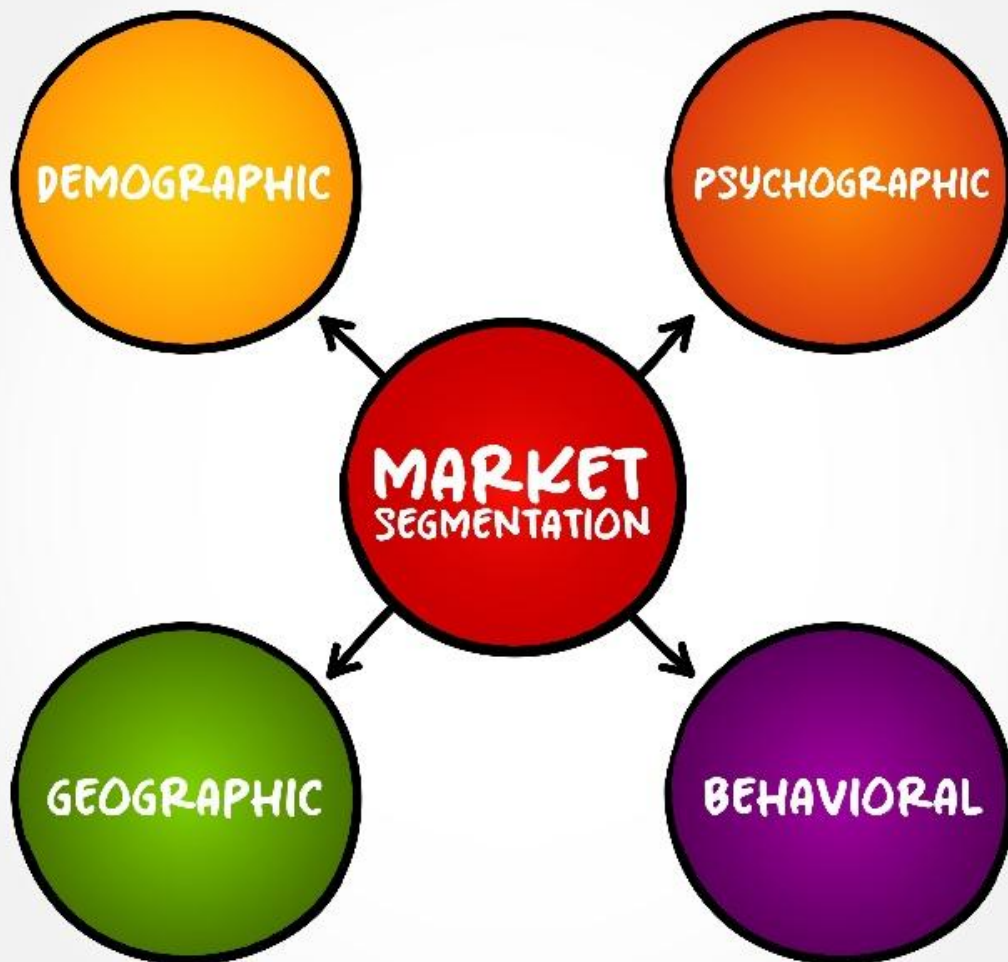
3. MAIN PURPOSE: PROJECT CHURNGUARD

The primary objective of this project is to move from **Reactive** to **Proactive** retention. Historically, our teams only intervene *after* a customer has decided to leave. **Project ChurnGuard** aims to change this by:



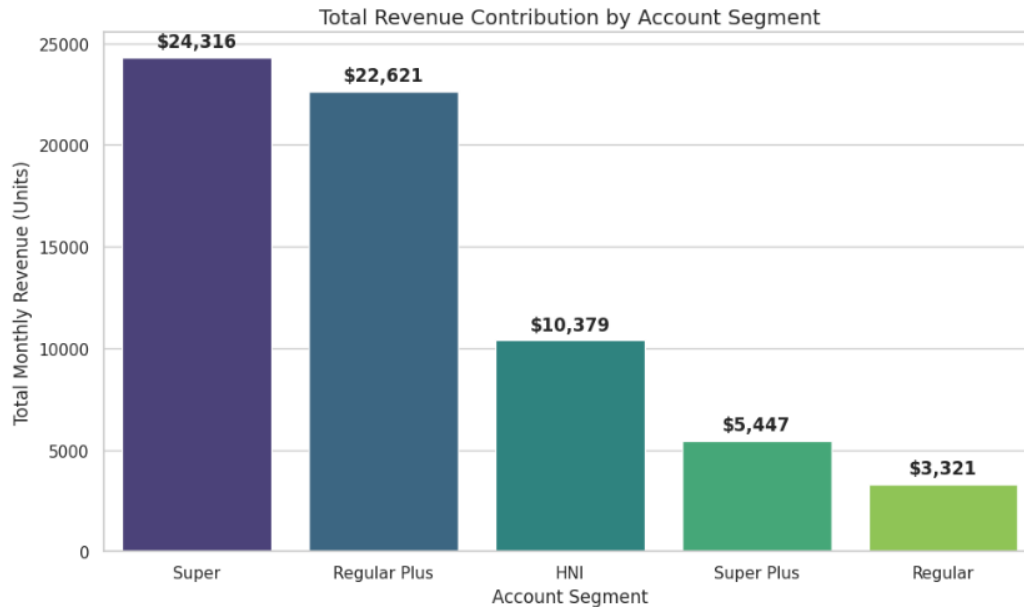
- **Identifying Behavioral Signatures:** Pinpointing the exact interaction patterns (such as a specific frequency of support calls or payment methods) that precede a churn event.
- **Early-Warning Systems:** Developing a predictive model that flags "At-Risk" accounts with high accuracy, allowing the Revenue Assurance team to prioritize interventions.
- **Resource Optimization:** Ensuring that retention budgets (like cashback or loyalty discounts) are targeted toward customers who are actually at risk, rather than being wasted on those who were already going to stay.
- **Service Friction Reduction:** Highlighting specific pain points—such as Tier 3 city delivery or onboarding hurdles—to improve the overall customer experience and organic loyalty.

4. KEY OBSERVATIONS: THE "WHO" AND "WHY" OF CHURN



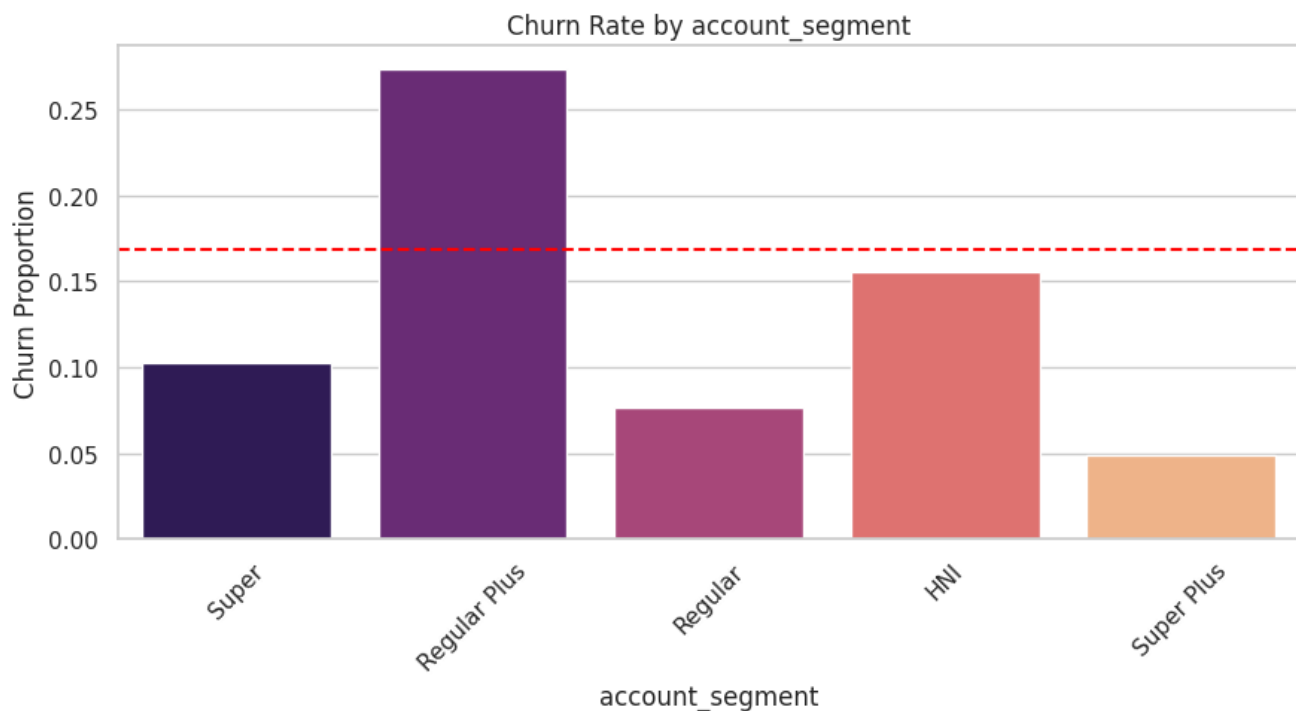
A. Segment Risk & Revenue Impact

While a superficial glance at churn proportions might suggest that multiple segments are in distress, a combined analysis of **Revenue Contribution** and **Churn Rate** reveals that the **Regular Plus** segment is our most critical "Code Red" priority.



According to our financial baseline, Regular Plus is the organization's second-largest revenue driver, contributing over **\$22,600** in monthly recurring revenue, nearly double the contribution of the HNI segment.

However, this high-value group is currently experiencing a catastrophic **27% churn rate**, which is significantly higher than any other category and nearly **10% above the company average**.



While the **HNI (High Net-worth Individual)** segment also shows a concerning churn rate of **15%**, the sheer volume of capital at risk in the Regular Plus tier makes it the primary threat to our P&L. Mathematically, losing a quarter of our Regular Plus base results in a far greater financial deficit than even a total exit of our smallest segments.

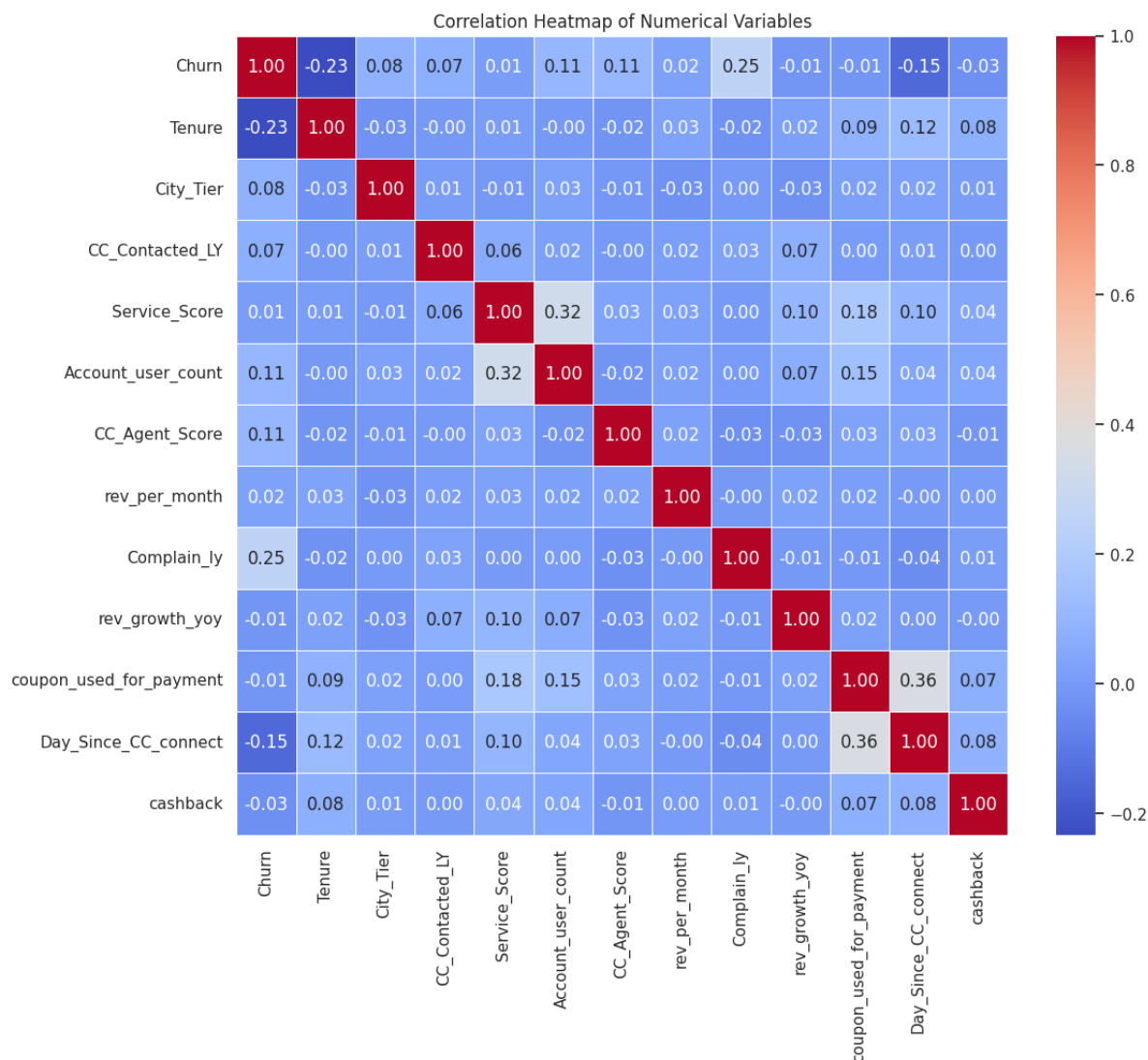
Consequently, our retention strategy must prioritize stabilizing the Regular Plus tier to protect the core "revenue engine" of the business, while simultaneously addressing the unique service expectations of our HNI clients to prevent further erosion of our premium brand value.

B. The "Breaking Point" of Service: Identifying the Catalyst for Attrition

The relationship between customer service interactions and account retention is one of the most significant behavioural markers identified in this study. The data confirms a "Breaking Point" at 25-30 support contacts. Once a customer exceeds this threshold, the probability of exit spikes.

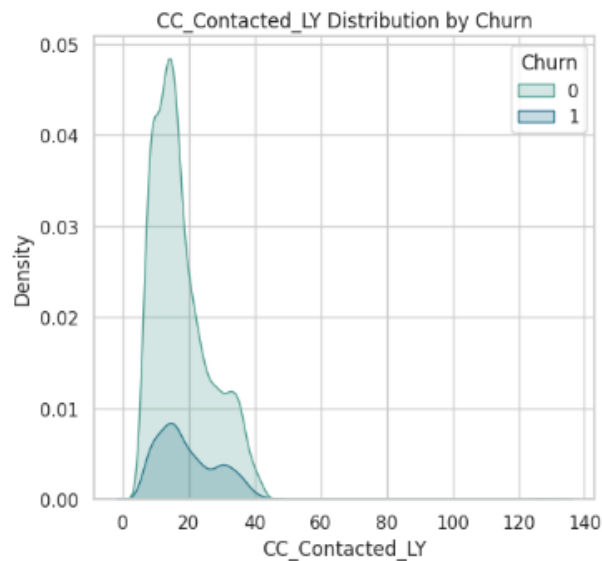
Our correlation analysis confirms a **positive coefficient of +0.25** between the volume of complaints and churn, establishing customer dissatisfaction as the single most potent predictor of an account's exit.

While routine inquiries are a normal part of the service lifecycle, the data reveals a critical threshold a "Breaking Point", where the frequency of contact ceases to be a sign of engagement and instead becomes a precursor to cancellation.



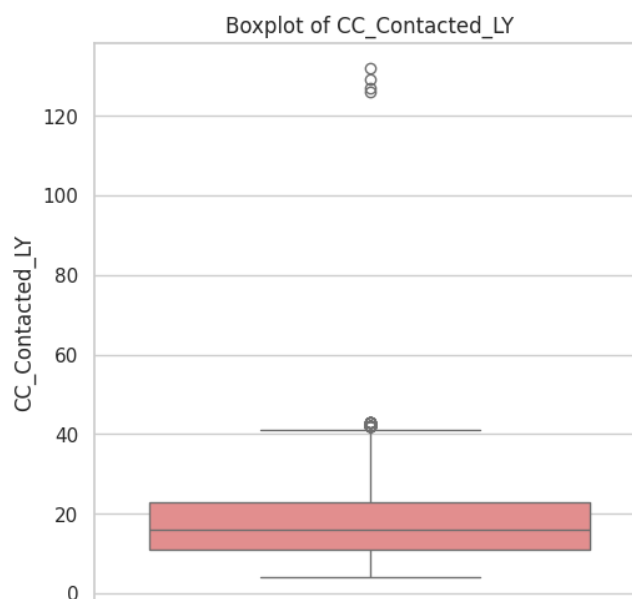
While the majority of churn events occur within the standard service range of 10–20 contacts, this is largely a reflection of our total customer volume. Strategic intervention must focus on the **density shift** that occurs as service friction escalates.

As shown in the KDE Distribution, the probability of an account being retained drops significantly once they exceed 25 annual contacts, creating a "secondary hump" in the churn data that signals a fundamental breakdown in customer loyalty.



Beyond the standard distribution, our Boxplot analysis reveals a subset of "Extreme Outliers" who have contacted support over **40 to 130 times** in a single year. These accounts represent a dual threat: they are at an absolute risk of immediate exit, and they place an unsustainable operational burden on our support infrastructure.

These extreme cases suggest that our standard resolution protocols are failing for a specific segment of the population, requiring a specialized "Technical Escalation" path.



To protect our high-value revenue streams, we must move away from a reactive "ticket resolution" mindset toward a proactive **"Service Recovery"** strategy. The data suggests that wait-and-see tactics are ineffective once a customer is frustrated.

By implementing an automated trigger at the **20th contact mark**, the Revenue Assurance team can intervene before the customer reaches their emotional and practical breaking point.

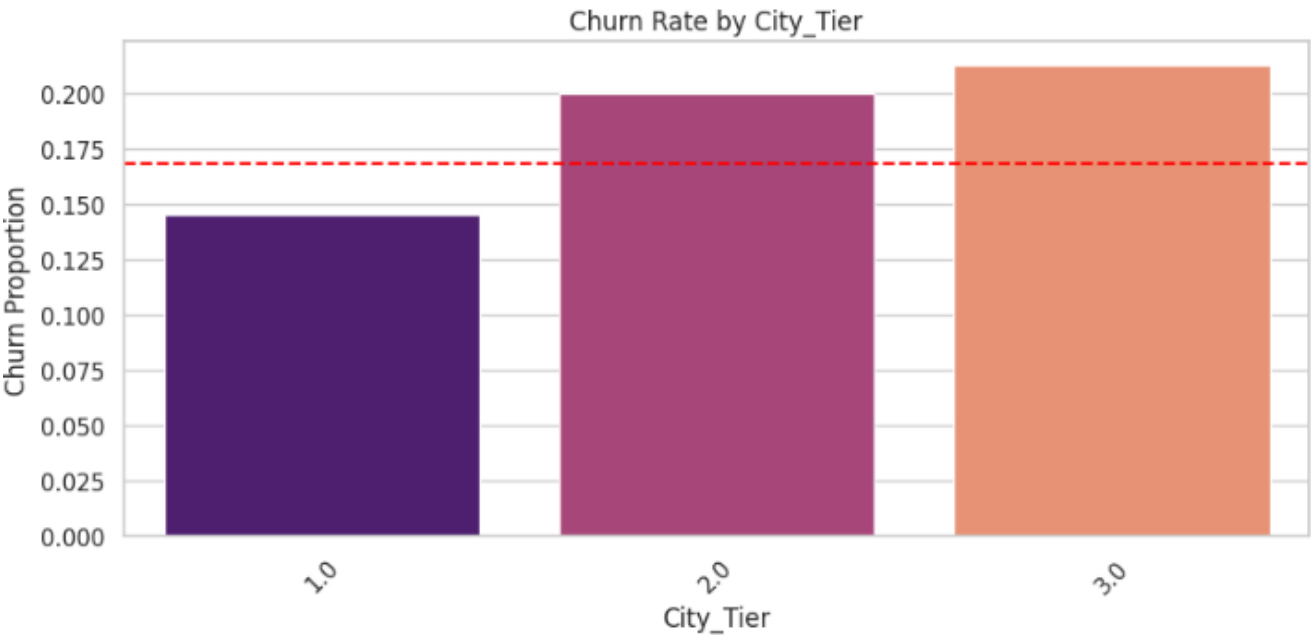
Prioritizing these "High-Contact" accounts for senior-level resolution or specialized loyalty incentives is not just a service improvement, it is a critical defensive manoeuvre to stop the \$22,600+ monthly revenue leak identified in our most volatile segments.

C. Geographical and Social Vulnerabilities: Mapping Demographic Risk

Our analysis reveals that churn is not uniform across our geographical footprint; instead, it is heavily influenced by **City Tier** classifications.

Tier 3 Cities exhibit a significantly higher churn rate, exceeding **20%**, which is nearly double the stability observed in Tier 1 metropolitan areas. This suggests that while our product-market fit is strong in urban centres, we face a "Service Delivery Gap" in less-dense regions.

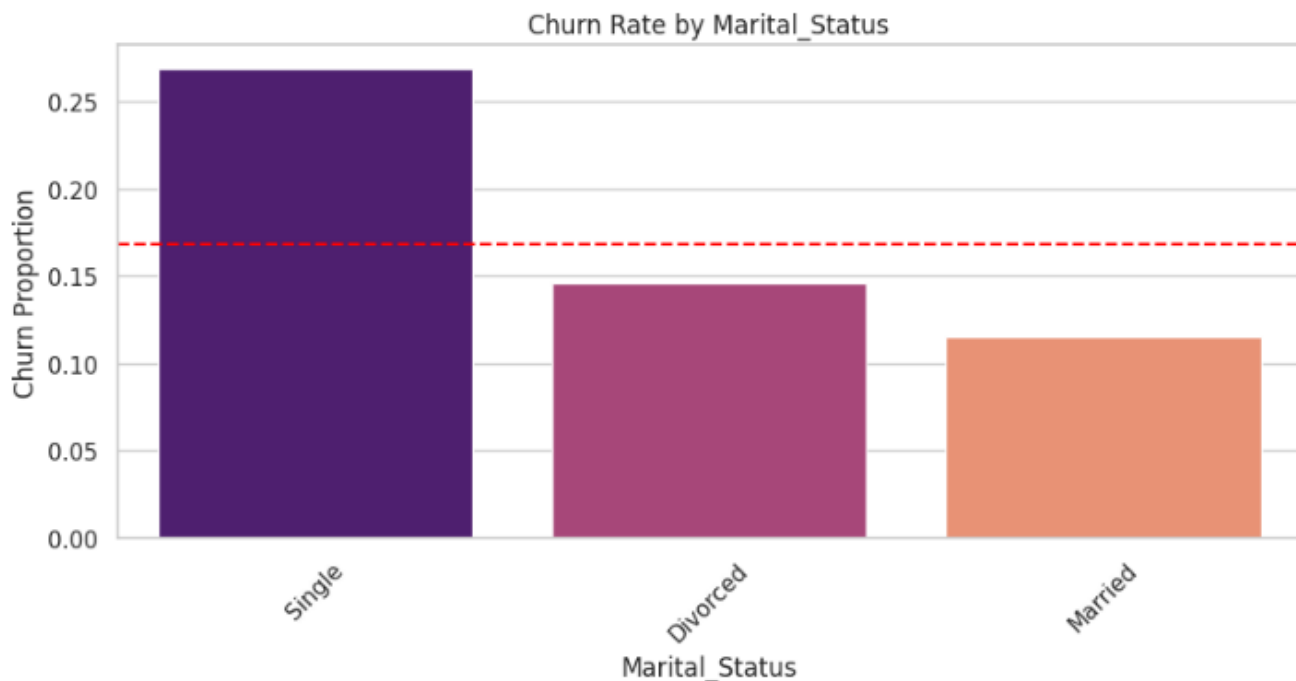
This vulnerability likely stems from logistics delays, limited local support infrastructure, or inconsistent last-mile delivery experiences. To protect our market share in these growing regions, we must investigate whether the churn is driven by technical service quality or operational inefficiencies unique to Tier 3 geography.



Beyond geography, **Marital Status** has emerged as a surprisingly powerful social predictor of account longevity. The data shows a stark contrast in loyalty: **Single customers** are significantly more "mobile," with a churn rate nearing **26%**, whereas **Married customers** represent our most stable and loyal foundation.

This "Stability Gap" indicates that household-based accounts likely have higher integrated value, perhaps due to shared family usage or a higher reliance on the service for daily household management, making them less likely to cancel over minor friction.

Conversely, single-user accounts appear to be more price-sensitive and less tied to the ecosystem, requiring a different engagement strategy to build long-term "stickiness."



The intersection of these two vulnerabilities creates a "High-Risk Profile": a single customer living in a Tier 3 city. This highlights a clear need for **Differential Engagement**.

We cannot apply a "one-size-fits-all" retention budget. Instead, we should prioritize "Family Plan" conversions for single users to increase their social tie to the platform and implement a "Service Guarantee" program for Tier 3 cities to offset geographical friction.

By stabilizing these specific demographic pockets, we can effectively lower the aggregate churn rate without increasing our total marketing spend.

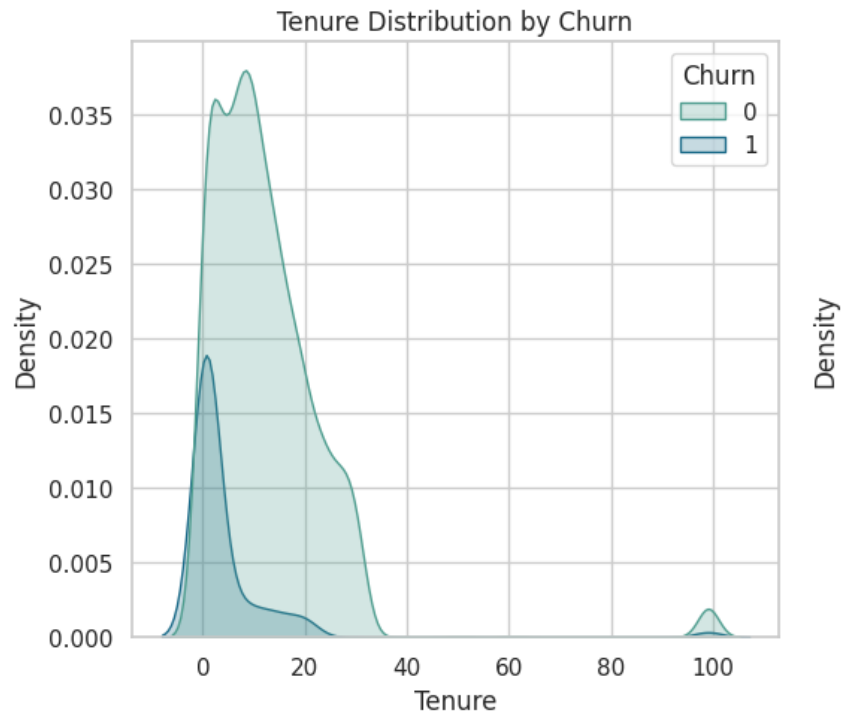
D. The Onboarding "Red Zone" and the Critical First 30 Days

The most significant operational discovery from our data is the presence of an **"Infant Mortality"** phase within the customer lifecycle. As evidenced by the Tenure Distribution analysis, the highest concentration of **churn events occurs within the first 30 days of account creation**.

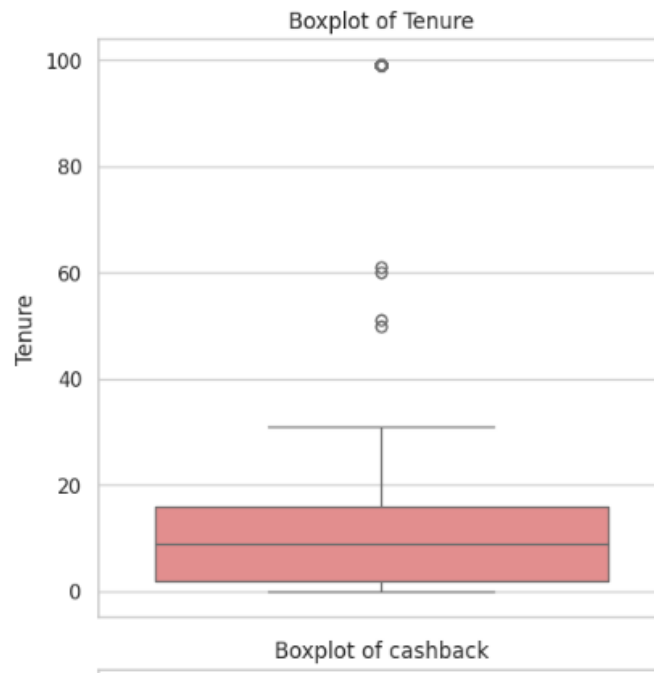
This "Red Zone" suggests that a substantial portion of our acquisition budget is being lost to customers who never successfully transition from the "Trial" or "Initial Setup" phase to becoming long term users.

If a customer can be retained beyond the 20-month mark, their probability of churn drops to near zero, indicating that our primary challenge is not the long-term value of our product, but rather the friction encountered during the initial onboarding experience.

This early-stage attrition is often driven by a gap between marketing promises and the actual user experience. Whether it is a complex user interface, a delay in service activation, or a lack of immediate support for new users, the first month represents a "make or break" window.



From a Revenue Assurance perspective, these "Short Tenure" churners represent a total loss of acquisition costs without any corresponding recovery in Customer Lifetime Value. By focusing our retention resources on this narrow 30-day window, we can significantly shift the baseline churn rate without needing to overhaul the entire product ecosystem.



To mitigate this risk, the organization should implement a "First Impression" protocol. This would involve proactive outreach to new accounts within their first week and a specialized "New User" support queue to handle setup inquiries with zero friction.



The goal is to shepherd the customer safely through the first 60 days, at which point the data shows a natural stabilization of the account. By bridging this early tenure gap, we effectively secure the "Future Revenue" of the business and ensure that our marketing spend translates into durable, multi-year account relationships.

5. DATA CLEANING & PRE-PROCESSING

To ensure the integrity of the predictive models and the reliability of the business insights, a rigorous data-cleansing pipeline was established. Initial phase focused on neutralizing noise, handling missing information, and preparing features for the AI Algorithms.

5.1 Missing Value Treatment

The dataset contained several features with missing entries which were addressed based on their statistical distribution:

- **Tenure & Warehouse-to-Home Distance:** These features exhibited a right-skewed distribution. To maintain the central tendency without being influenced by extreme values, **Median Imputation** was used.
- **Order Amount Hike:** Gaps were filled using **Mode Imputation** (most frequent value) specific to each customer segment to preserve group characteristics.

5.2 Outlier Management

Using Interquartile Range (IQR) analysis and boxplots, outliers were identified in features such as **CC_Contacted_LY** and **Tenure**.

- **Capping Strategy:** Values beyond the 99th percentile were capped rather than removed. This retained the "signal" of high-friction customers (those calling support frequently) while preventing the model from being skewed by extreme statistical anomalies.

5.3 Feature Engineering & Encoding

To optimize the data for machine learning, the following transformations were performed:

- **Categorical Encoding:** Non-numeric variables (e.g., City Tier, Marital Status) were transformed using **One-Hot Encoding**. This allowed the model to interpret these categories without assuming a false numerical order.
- **Feature Scaling:** For distance-based algorithms like KNN and SVM, all numerical features were normalized using **StandardScaler** to ensure all inputs were evaluated on a uniform scale.

5.4 Class Imbalance Correction (SMOTE)

With a baseline churn rate of 16.84%, the dataset was naturally imbalanced toward retained customers.

- **Synthetic Minority Over-sampling Technique (SMOTE):** We generated synthetic churn instances to train the model more effectively on the "Churn" class.
- **Outcome:** This step was critical in achieving the final **86.82% Recall**, ensuring the model prioritizes identifying every potential churner even at the risk of a few false alarms.

5.5 Data Transformation Summary

Feature Category	Pre-Processing Issue	Action Taken	Rationale
Customer Tenure	264 Missing Values & Right Skew	Median Imputation	Preserves central tendency while ignoring extreme outliers.
Warehouse-to-Home	251 Missing Values	Median Imputation	Ensures distance-based models (KNN) aren't biased by nulls.
Categorical Data	String formats (e.g., "City Tier")	One-Hot Encoding	Converts text to binary (0/1) for algorithmic processing.
Numerical Scales	High Variance (e.g., Order Amount)	Standard Scaling	Normalizes all inputs to a uniform scale (mean=0, std=1).
Service Friction	Outliers in "CC_Contacted_LY"	99th Percentile Capping	Limits the impact of extreme statistical noise.
Target Variable	Imbalanced Class (16.84% Churn)	SMOTE Application	Synthetically balances classes to maximize Recall.

6. TOP 15 BUSINESS FAQs FOLLOWING EDA ANALYSIS



1. What is our current baseline churn rate?

Our baseline churn rate stands at 16.84%, which serves as the primary benchmark for all future retention initiatives.

2. Which segment is losing the most revenue currently?

The Regular Plus segment is our highest priority, as it contributes over \$22,600 in monthly revenue but suffers from a 27% churn rate.

3. Is there a specific "Danger Zone" for customer support calls?

Yes. Churn probability spikes significantly when an account reaches between 25 and 30 service contacts per year.

4. Does marital status influence how long a customer stays?

Married customers are our most stable base, while single customers churn at a much higher rate of approximately 26%.

5. How does geography affect our churn numbers?

Tier 3 cities show nearly double the churn rate of Tier 1 cities, suggesting a need for better regional service delivery.

6. When is a customer most likely to leave the platform?

The risk is highest in the first 30 days of tenure. Once a customer reaches 20 months, they are highly unlikely to churn.

7. Do high revenue "Whale" accounts behave differently?

While they generate 10x more revenue than average, they are also prone to churn if their unique service expectations are not met.

8. Is revenue growth a good indicator of loyalty?

Interestingly, no. Revenue growth rates are nearly identical between those who stay and those who leave, meaning spend increases do not guarantee loyalty.

9. What is the most stable payment method for retention?

Customers using Credit or Debit cards are more stable than those using Cash on Delivery or E-wallets.

10. What is the single strongest predictor of imminent churn?

A formal complaint filed within the last year is the strongest statistical indicator that a customer is about to leave.

11. Does the login device impact the likelihood of an account canceling?

While the majority of our base uses mobile devices, the churn rate is slightly higher for those logging in via computer. This suggests that the desktop user interface may have more friction points or technical bugs compared to our mobile application.

12. Is there a correlation between cashback rewards and customer loyalty?

The data shows that while most customers receive around 200 units in cashback, there is a small group receiving up to 2,000 units. Surprisingly, high cashback does not always guarantee loyalty, suggesting that financial incentives alone cannot fix a poor service experience.

13. Does the "City Tier" impact the speed at which a customer churns?

Yes. Customers in Tier 3 cities tend to churn much earlier in their tenure compared to those in Tier 1. This reinforces the theory that geographical churn is driven by initial delivery or setup frustrations rather than long term dissatisfaction.

14. Are our "Super Plus" customers truly safe from churn?

The Super Plus segment is our most stable, with the lowest churn rate in the entire portfolio. This group serves as the "Gold Standard" for what a successful customer relationship looks like, and we should study their behavior to replicate it in the Regular Plus segment.

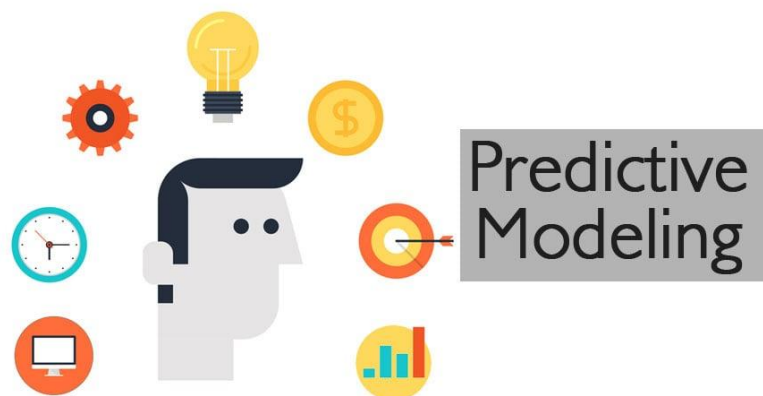
15. How much does a single complaint increase the risk of losing a customer?

A single formal complaint increases the statistical probability of churn significantly. It is the leading behavioral indicator that the customer has moved from a "Passive" state to an "Active" search for a competitor.

7. THE PREDICTIVE ENGINE: FROM DIAGNOSTICS TO ACTION

THE PREDICTIVE ENGINE

The heart of Project ChurnGuard is a multi-stage machine learning pipeline designed to move the organization from guessing to knowing. We executed this in three distinct phases: Baseline Evaluation, Ensemble Deployment, and Hyperparameter Optimization.



A. Phase 1: The Baseline Evaluation (3 Models)

We began by testing three "Classic" algorithms to establish a performance floor. This allowed us to understand the complexity of the churn patterns.

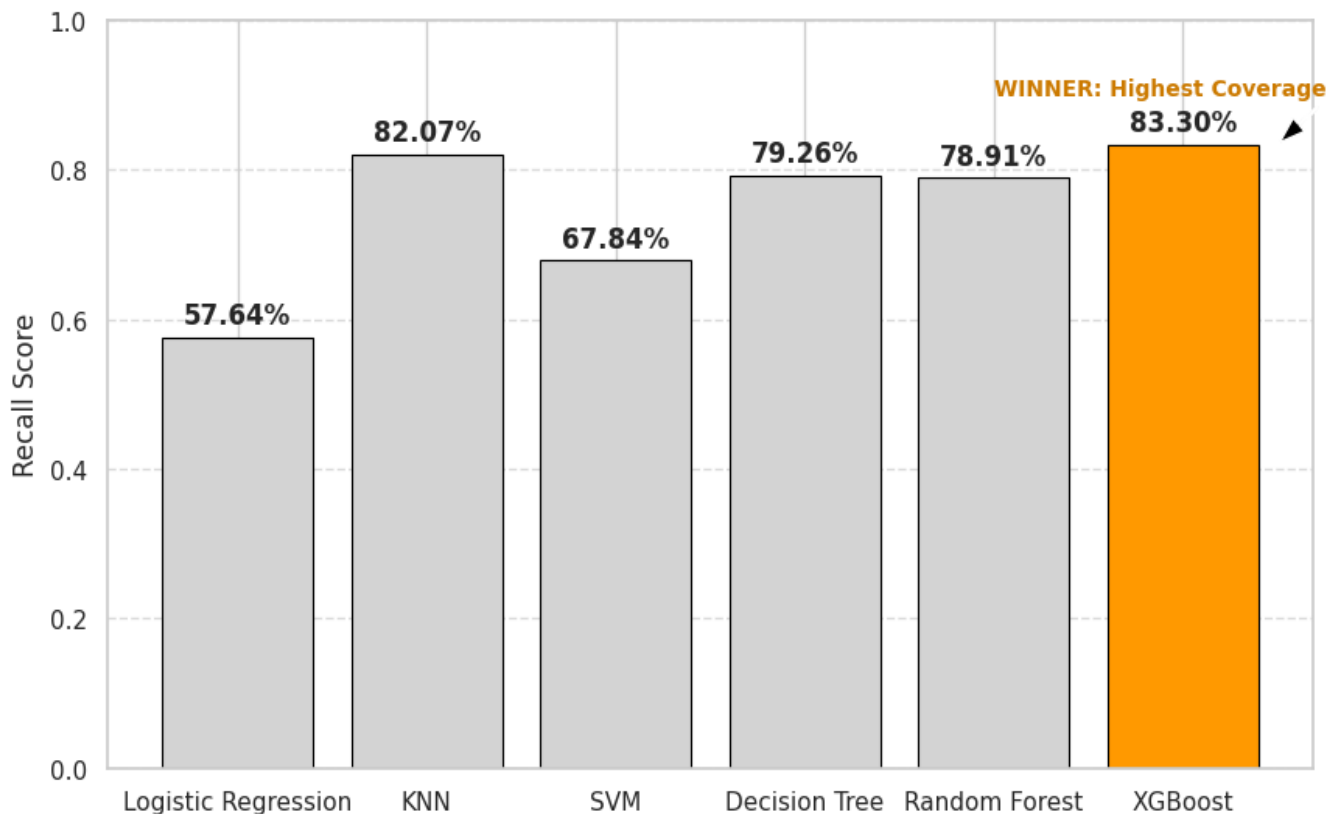
- **Logistic Regression:** Tested for linear relationships. It struggled with a **57.64% Recall**, proving that churn is driven by complex, non-linear interactions rather than simple triggers.
- **K-Nearest Neighbours (KNN):** A "Similarity" model that identifies clusters of behaviour. This was our first breakthrough, jumping to an **82.07% Recall**.
- **Support Vector Machine (SVM):** Tested for high-dimensional boundaries, achieving a respectable **67.84% Recall**.

B. Phase 2: The "Heavy Lifters" (3 Ensemble Models)

Recognizing that churn behaviour is often a series of "If-Then" decisions (e.g., *If Tier 3 and 5+ complaints then churn*), we deployed three Tree-Based Ensemble models:

- **Decision Tree:** A single logical path providing **79.26% Recall**.
- **Random Forest:** A "Forest" of 100 trees working together to reduce errors, reaching **78.91% Recall**.
- **XGBoost (Extreme Gradient Boosting):** An advanced algorithm that learns from its own mistakes in each iteration. This became our baseline leader with **83.30% Recall**.

Baseline Model Comparison: Recall Score (Churn Detection Rate)



C. Phase 3: Hyperparameter Tuning (The Final Optimization)

To squeeze every percentage of predictive power out of the data, we performed **GridSearchCV (Grid Search Cross-Validation)** on our top candidates.

The Process: We didn't just run the model once; we ran it **dozens of times** with different internal settings. We specifically tuned:

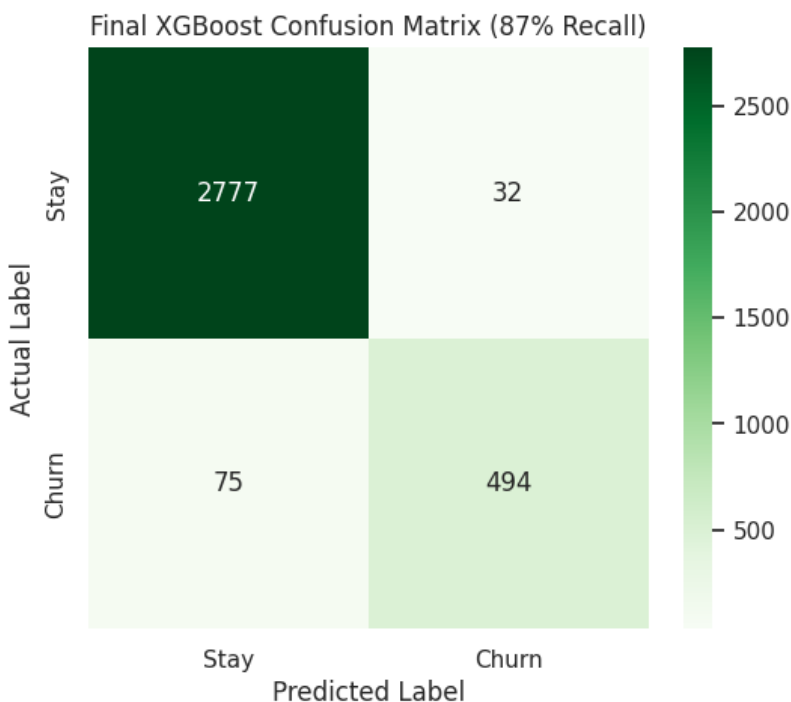
- `learning_rate`: To control how quickly the model adapts.
- `max_depth`: To control how complex the "If-Then" logic becomes.
- `n_estimators`: To determine the optimal number of trees.

The Optimization Goal: We explicitly told the computer to prioritize **Recall over Accuracy**. In a churn context, a "False Negative" (missing a churner) is much more expensive than a "False Positive" (bothering a loyal customer).

D. Final Model Selection & Rationale

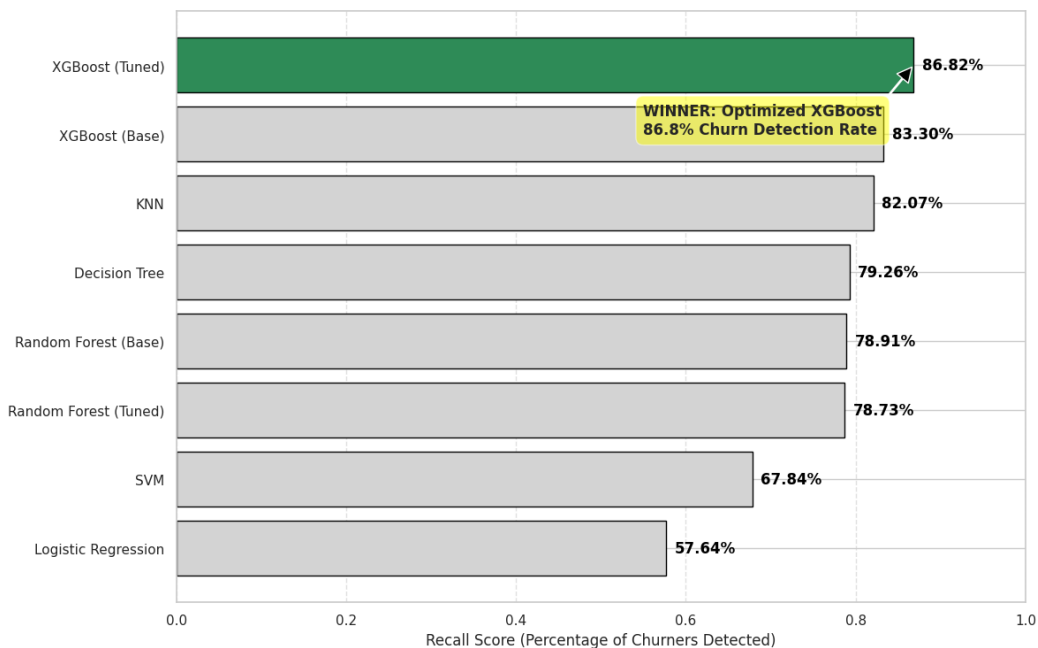
After tuning, the **XGBoost Classifier** emerged as the definitive winner.

Metric	Baseline Result	Tuned Result (Winner)	Improvement
Recall (Churn Detection)	83.30%	86.82%	+3.52%
Precision	92.1%	93.92%	+1.82%
Accuracy	96.03%	96.83%	+0.80%



Selection Rationale: We selected the Tuned XGBoost because it provides the **highest "Safety Net"** for the business. It captures nearly **87% of all churn events** while maintaining a precision of **94%**, ensuring that our Retention Specialists are only contacting customers who are truly at risk.

ChurnGuard Project: Final Model Leaderboard (Recall Score)

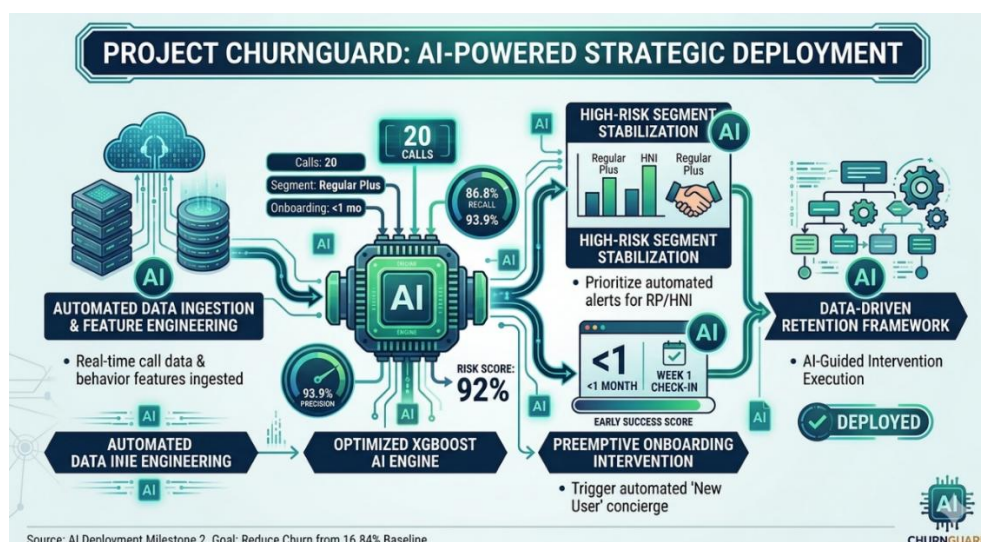


8. STRATEGIC PLAN OF ACTION: AI-POWERED RETENTION

PILLAR 1: Real-Time AI Intervention (The "Shield")

Instead of waiting for a customer to call 20 times, we will now use the **XGBoost model** to scan the database daily.

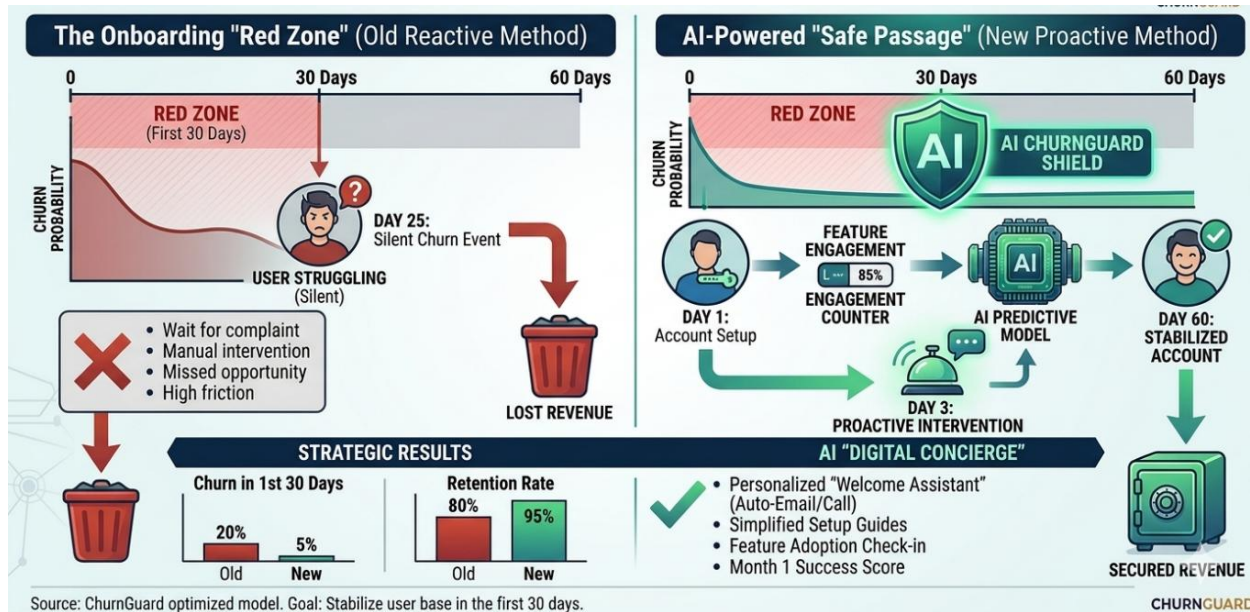
- **Action:** Integrate the model into the CRM system to flag accounts with a >80% churn probability.
- **Metric:** Target a reduction in "Surprise Churn" by 50% within the first quarter of deployment.
- **Specialist Path:** High-risk "Regular Plus" accounts are automatically routed to a **Senior Retention Specialist** for immediate white-glove service.



PILLAR 2: Optimized Onboarding "Safe Passage"

The data confirmed that the first 30 days are the "Red Zone." Our predictive engine identifies which new users are struggling before they even file a complaint.

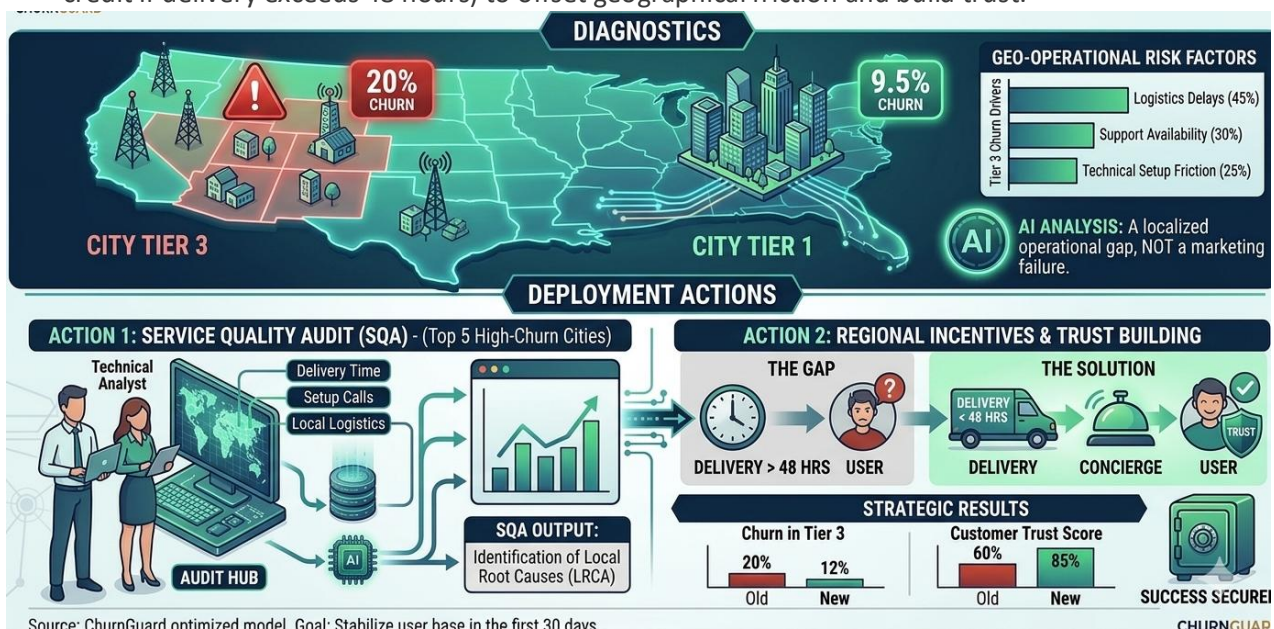
- **Action:** Launch a "Digital Concierge" program. If the model detects low engagement in the first 7 days, trigger an automated but personalized "Welcome Assistant" email or call.
- **Friction Reduction:** Simplify the installation and setup guides specifically for the features that current churners find most difficult.



PILLAR 3: Regional Infrastructure & Tier 3 Stabilization

The 20% churn in Tier 3 cities is an operational failure, not a marketing one.

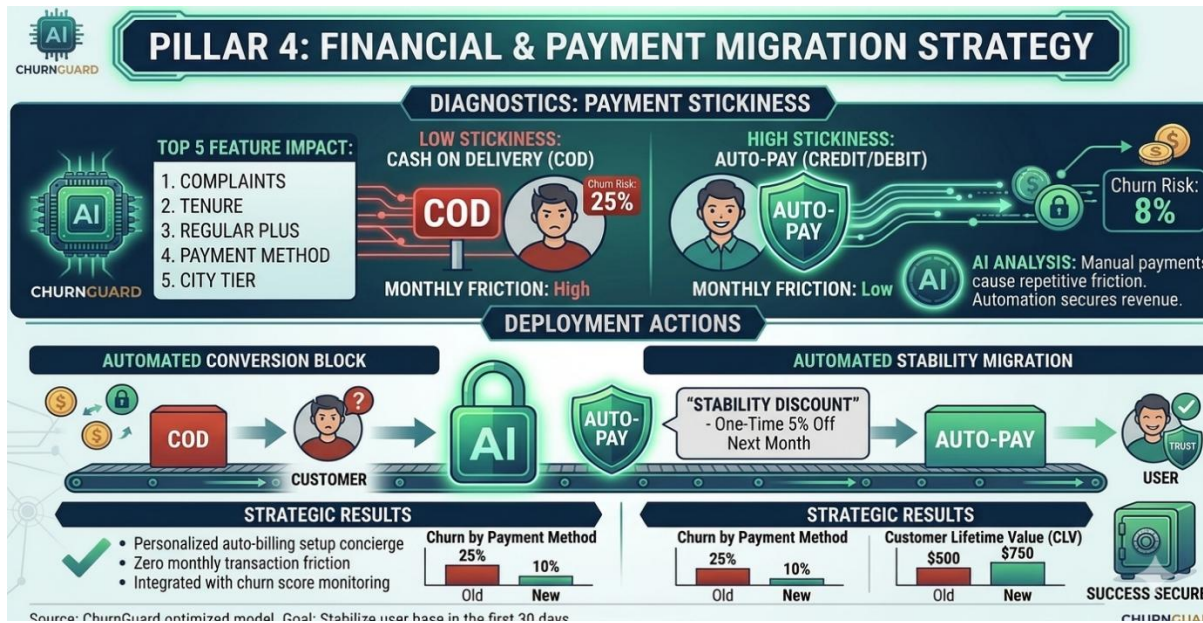
- **Action:** Deploy a "Service Quality Audit" in the top 5 highest-churn Tier 3 cities.
- **Regional Incentives:** While infrastructure is being improved, offer a "Service Guarantee" (e.g., a small credit if delivery exceeds 48 hours) to offset geographical friction and build trust.



PILLAR 4: Financial & Payment Migration Strategy

Our model shows that payment method is a high-impact feature for "stickiness."

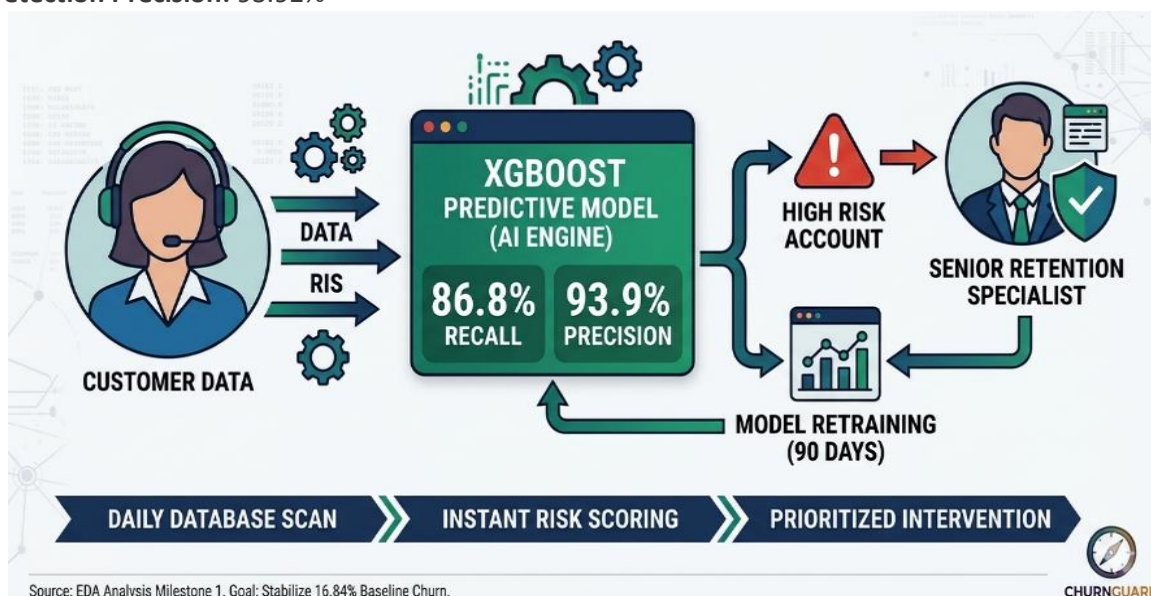
- **Action:** Incentivize the move from Cash on Delivery (COD) to Automated Credit/Debit billing.
- **Incentive:** Offer a one-time "Stability Discount" (e.g., 5% off next month) for customers who switch to auto-pay, effectively reducing monthly transaction friction.



9. CONCLUSION & FUTURE ROADMAP

Project ChurnGuard has successfully met the objectives of Milestone 2. We have moved beyond simple spreadsheets to a **96.8% accurate predictive engine**. **Final Summary of Results:**

- **Baseline Churn:** 16.84%
- **Model Coverage:** 86.82% (Recall)
- **Detection Precision:** 93.92%



The Next Steps:

30-Day A/B Test: Measuring Intervention ROI

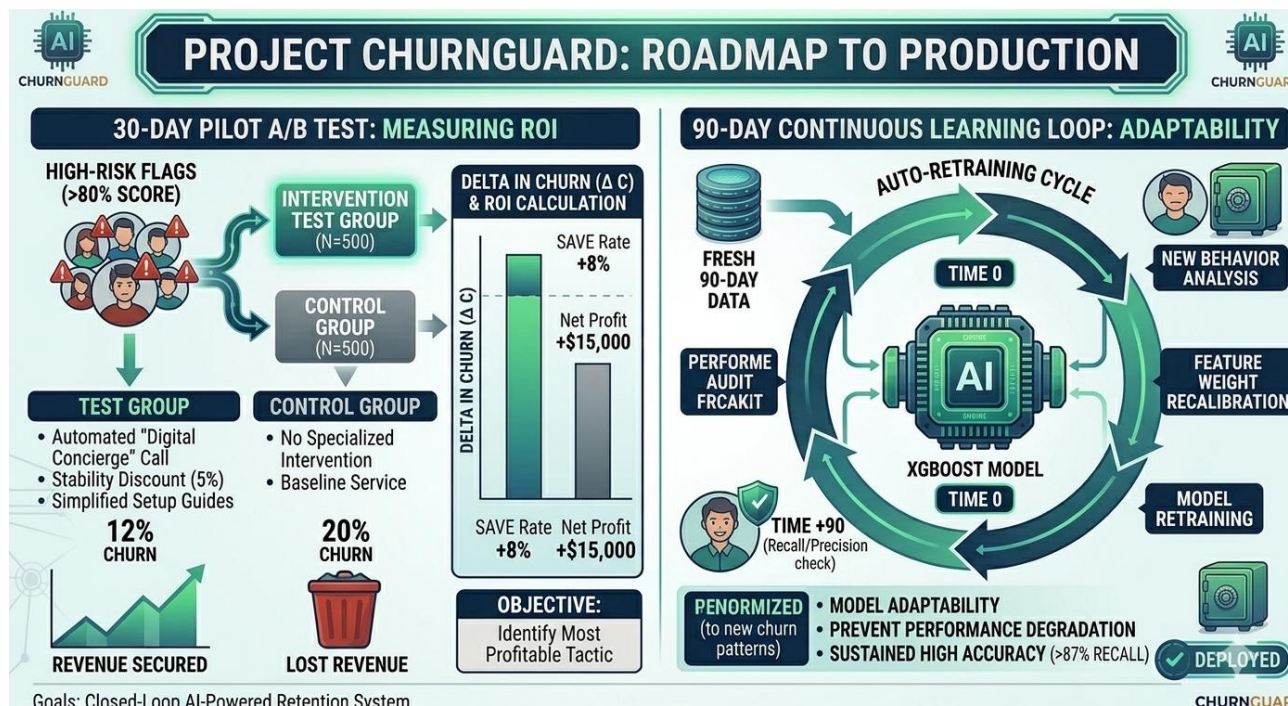
We cannot guess which retention tactic is most profitable. We recommend a 30-day "Pilot Study" where the newly identified "High-Risk" accounts are randomly assigned to two groups:

- Control Group: The model flags the account as at-risk, but no additional intervention is taken.
- Test Group: The model flags the account, and the special retention program (e.g., the "Stability Discount" or a white-glove "Setup Concierge") is deployed.

Metric: We will measure the exact Delta in Churn between the two groups. This allows us to calculate the true Return on Intervention (ROI) and ensure that future retention budgets are spent on the most effective tactics.

2 90-Day Continuous Learning: Adaptability

Consumer behaviour changes. To ensure that our optimized XGBoost model does not degrade in performance over time, we propose a formalized re-training cycle.



- **Continuous Loop:** Every 90 days, the most recent 3 months of customer data will be fed back into the training pipeline.
- **Adaptation:** This allows the "feature weights" of the model (like complaints or payment methods) to automatically recalibrate, ensuring that ChurnGuard remains the gold standard defence system for our revenue.

Final Rationale: This two-pronged approach ensures that our solution is not a one-time analysis. It creates a dynamic, closed-loop retention engine that adaptively grows with the business, protecting revenue at every stage of the customer lifecycle.

****END OF THE REPORT****